

In-Class Worksheet

Data Visualization and R Programming

Name: _____

Date: _____

Course: H524 Introduction to Biostatistics

Topic: Data Presentation, Histograms, Boxplots, Introduction to R Programming

1 Part 1: Review and Setup

1.1 Exercise 1.1: Quick Review

From previous material, what were the three measures of central tendency for our blood pressure data `c(170, 193, 110, 135, 135, 115)`?

- Mean: _____ mmHg
- Median: _____ mmHg
- Mode: _____ mmHg

```
1 # Load the data again
2 blood_pressure <- c(170, 193, 110, 135, 135, 115)
3
4 # Verify your answers
5 mean(blood_pressure)
6 median(blood_pressure)
```

2 Part 2: Introduction to Data Visualization

2.1 Exercise 2.1: Creating Histograms

We'll work with simulated guinea pig survival data from three groups:

```
1 # Guinea pig survival data (tuberculosis study)
2 control <- c(312, 345, 287, 398, 421, 356, 389, 334, 367, 401)
3 low_dose <- c(289, 267, 301, 234, 198, 245, 278, 213, 256, 298)
4 high_dose <- c(156, 189, 134, 167, 123, 145, 178, 201, 143, 134)
5
6 # Create histograms
7 hist(control, main="Control_Group_Survival", xlab="Days", col="lightblue")
8 hist(low_dose, main="Low_Dose_Group_Survival", xlab="Days", col="
  lightgreen")
9 hist(high_dose, main="High_Dose_Group_Survival", xlab="Days", col="
  lightcoral")
```

Key Questions:

1. What do you notice about the shape of each distribution?
2. Which group appears to have the best survival prospects?
3. Why is it important to use the same x-axis scale when comparing groups?

2.2 Exercise 2.2: AI-Enhanced Histogram Creation

AI Prompt: “Help me create side-by-side histograms for comparing three groups of guinea pig survival data. The data should use the same scales and bin widths for fair comparison.”

AI Response (fill in):

```
1 # Paste AI-generated code here:
```

Verification Checklist:

- ☐ Same x-axis scale across all histograms?
- ☐ Same number of bins?
- ☐ Clear labels and titles?
- ☐ Different colors for each group?

3 Part 3: Boxplots for Group Comparison

3.1 Exercise 3.1: Understanding the Five-Number Summary

For the control group data, calculate the five-number summary:

```
1 control <- c(312, 345, 287, 398, 421, 356, 389, 334, 367, 401)
2
3 # Calculate five-number summary
4 summary(control)
5
6 # Individual components
7 min(control)
8 quantile(control, 0.25) # Q1
9 median(control)        # Q2
10 quantile(control, 0.75) # Q3
11 max(control)
```

Fill in the five-number summary:

- Minimum: _____
- Q1 (25th percentile): _____
- Q2 (Median): _____
- Q3 (75th percentile): _____
- Maximum: _____

3.2 Exercise 3.2: Creating Boxplots

```
1 # Create individual boxplots
2 boxplot(control, main="Control_Group", ylab="Survival_Days")
3 boxplot(low_dose, main="Low_Dose_Group", ylab="Survival_Days")
4 boxplot(high_dose, main="High_Dose_Group", ylab="Survival_Days")
5
6 # Side-by-side comparison
7 boxplot(control, low_dose, high_dose,
8         names=c("Control", "Low_Dose", "High_Dose"),
9         main="Guinea_Pig_Survival_by_Treatment_Group",
10        ylab="Survival_Days",
```

```
11 col=c("lightblue", "lightgreen", "lightcoral"))
```

Analysis Questions:

1. Which group has the highest median survival?
2. Which group shows the most variability (largest IQR)?
3. Are there any potential outliers visible?

4 Part 4: R Programming Fundamentals

4.1 Exercise 4.1: Working with Vectors and Data Frames

```
1 # Create a data frame combining all groups
2 survival_data <- data.frame(
3   days = c(control, low_dose, high_dose),
4   group = c(rep("Control", length(control)),
5             rep("Low_Dose", length(low_dose)),
6             rep("High_Dose", length(high_dose)))
7 )
8
9 # Explore the data frame
10 str(survival_data)
11 head(survival_data)
12 summary(survival_data)
```

4.2 Exercise 4.2: Basic Data Manipulation

```
1 # Calculate group means
2 aggregate(days ~ group, data = survival_data, FUN = mean)
3
4 # Calculate group standard deviations
5 aggregate(days ~ group, data = survival_data, FUN = sd)
6
7 # Create a subset for just the control group
8 control_subset <- subset(survival_data, group == "Control")
```

Fill in the results:

- Control group mean: _____
- Low dose group mean: _____
- High dose group mean: _____

5 Part 5: AI-Enhanced Analysis Challenge

5.1 Exercise 5.1: Complete Statistical Summary

AI Challenge: Use AI to help create a comprehensive analysis comparing the three treatment groups.

Your AI Prompt: "I have guinea pig survival data for three groups: control, low dose, and high dose. Help me create a complete statistical comparison including descriptive statistics, appropriate visualizations, and interpretation of the results."

AI Response Documentation:

1. Descriptive Statistics Code:

```
1 # Paste AI code here:
```

2. Visualization Code:

```
1 # Paste AI code here:
```

3. AI Interpretation: (Write the AI's interpretation of the results)

Your Verification:

- ☐ Are the statistical calculations correct?
- ☐ Do the visualizations effectively show group differences?
- ☐ Is the interpretation scientifically sound?
- ☐ What would you add or modify?

6 Part 6: Connecting to Biostatistics Concepts

6.1 Exercise 6.1: Population vs. Sample Discussion

Questions for Discussion:

1. In this guinea pig study, what is the population of interest?
2. What is our sample?
3. How might these results generalize to the broader population?
4. What are potential limitations of this study design?

6.2 Exercise 6.2: Setting Up for Statistical Inference

In upcoming classes we'll learn about probability. Based on today's data:

1. If we randomly selected one guinea pig from the control group, would you expect its survival to be exactly the mean survival time? Why or why not?
2. How confident are you that the control group truly has better survival than the high dose group? What additional information would help you be more certain?

7 Wrap-up and Preview

Key Takeaways:

- Histograms show the shape and distribution of data
- Boxplots efficiently compare groups using five-number summaries
- R data frames organize different types of data together
- AI can accelerate analysis but requires human verification

Next Class: We'll dive deeper into numerical summary measures, rates and standardization, and advanced R programming techniques.

8 AI Usage Documentation

Document your AI interaction:

- AI tools used: _____
- Most helpful AI suggestion: _____
- Code that needed correction: _____
- Verification methods used: _____
- Overall AI effectiveness (1-10): _____